

Noah Kim

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EDUCATION

Stony Brook University

Aug 2025 - Present

B.E. Mechanical Engineering and B.S. Applied Mathematics — Stony Brook, NY

GPA: 3.75 | Expected graduation: May 2029

The High School for Mathematics, Science, and Engineering at CCNY

2021 - 2025

Advanced Regents Diploma with Honors, 4.00 GPA — New York, NY

AP Scholar with Distinction; CCNY Dual Enrollment; President's Award; Seal of Service; Varsity Letter

PROFESSIONAL EXPERIENCE

Stony Brook University

Aug 2025 - Present

Engineering Research Associate — Dr. Eric Josephs

- Derive governing equations and implement finite element analysis (FEA) routines in Python that generate robust auxetic lattices and map complex cell geometry into arbitrary three-dimensional volumes.
- Achieved over 70% redirection of axial impact load into transverse load paths versus a carbon fiber reinforced polymer (CFRP) crush baseline, validated in both simulation and physical crush testing.
- First author on a manuscript establishing the physical realizability and engineering applicability of these auxetic geometries; currently under review.

New York City Department of Transportation

Jul 2024 - Jun 2025

Research Assistant — New York, NY

- Built high-fidelity simulations of Bell 429 VTOL rotary mechanisms to evaluate structural integrity and rotor efficiency.
- Conducted stress and ultimate-load analysis to identify critical failure points and verify that aerodynamic components met safety and performance benchmarks.
- Synthesized simulation results into risk-reward frameworks for department leadership, informing decisions on the cost-effectiveness and operational viability of proposed rotorcraft configurations.

EXTRACURRICULAR EXPERIENCE

CubeSat

Aug 2026 - Present

- Designing and manufacturing the attitude determination and control system (ADCS) and thermal control system (TCS) for a 2U low-Earth-orbit (LEO) satellite, targeting a Spring 2027 launch.
- Preparing ground-station tracking and telemetry analysis procedures for post-launch operations.

PROJECTS

LiDAR Survey Drone

Jan 2026 - Present

- Designing a fully remote-controlled drone airframe with autonomous VTOL capability and an onboard LiDAR payload targeting 2,560 measured points per second at nominal survey altitude.
- Built on a Raspberry Pi Zero 2 W (quad-core 64-bit ARM Cortex-A53, 1 GHz) with a CSI camera module; developing a 2.4 GHz video link to a head-mounted receiver.

TECHNICAL SKILLS

CAD & Modeling: SolidWorks, Autodesk Inventor, Fusion 360, Revit

Simulation & Analysis: FEniCS, finite element analysis (FEA)

Programming: Python, R, MATLAB, HTML

Hardware & Embedded: Arduino, Raspberry Pi, Bluetooth (HC-05) serial communication, CSI camera integration